

Ex.No:2a	Scheduling of project with PERT techniques to estimate the completion time
Date:	

AIM:

To schedule a project using the PERT (Program Evaluation and Review Technique) to estimate the expected project completion time.

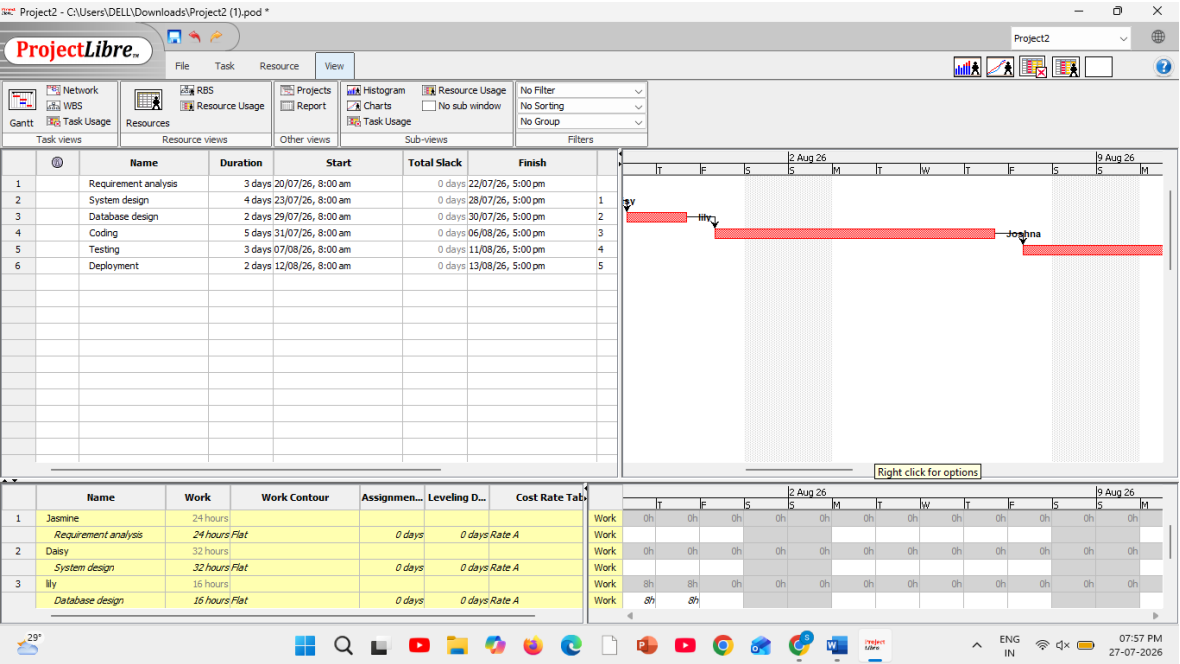
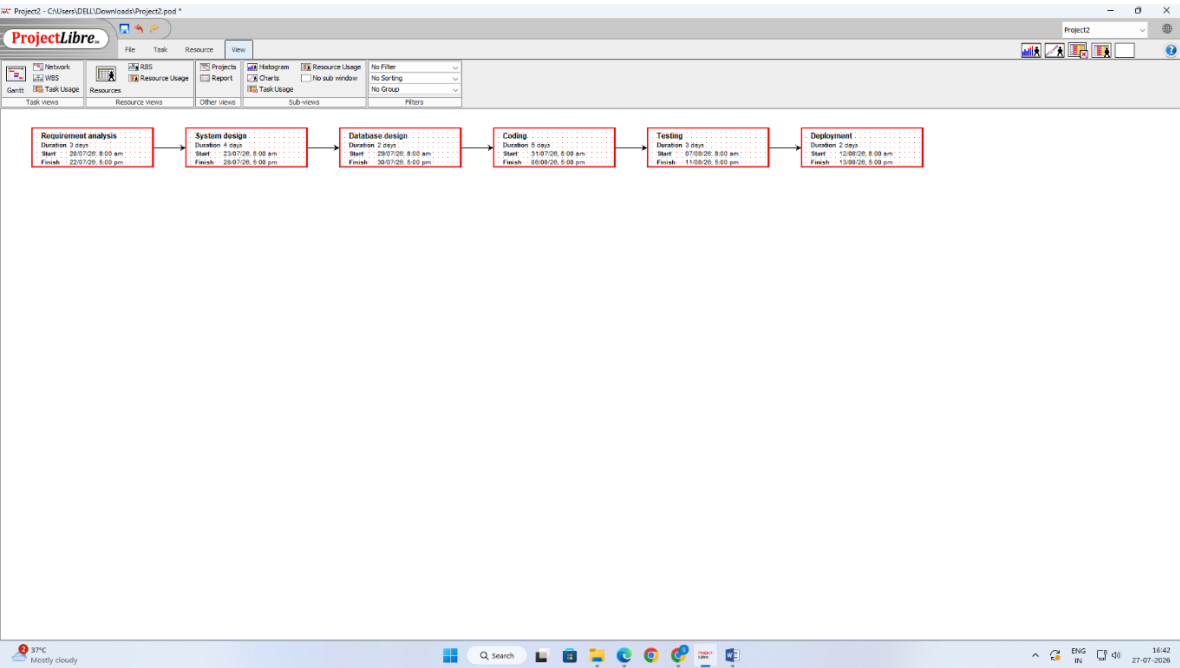
ALGORITHM:

1. List all project activities and their sequence.
2. Estimate the optimistic (O), most likely (M), and pessimistic (P) time for each activity.
3. Calculate the expected time using $TE = (O + 4M + P) / 6$.
4. Construct the PERT network diagram.
5. Identify all possible paths from start to finish.
6. Determine the critical path by finding the longest path.
7. Calculate the total expected project completion time

PROCEDURE:

1. Open a project scheduling tool or draw a PERT network manually.
2. Enter all project activities and their dependencies.
3. Assign optimistic, most likely, and pessimistic time estimates.
4. Compute the expected time for each activity.
5. Create the PERT network connecting all activities.
6. Identify the critical path and calculate the project duration.
7. Analyze the results and generate the project schedule.

OUTPUT:



RESULT:

Ex.No:2b	Scheduling of project with CPM echniques to estimate the completion time
Date:	

AIM:

To schedule a project using the Critical Path Method (CPM) and determine the critical path and project completion time.

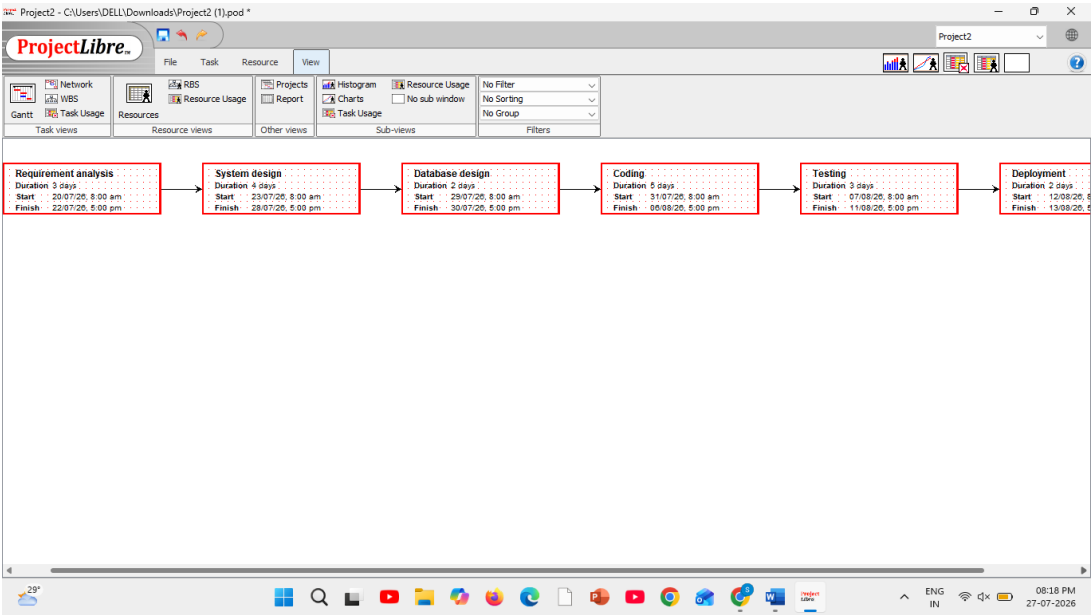
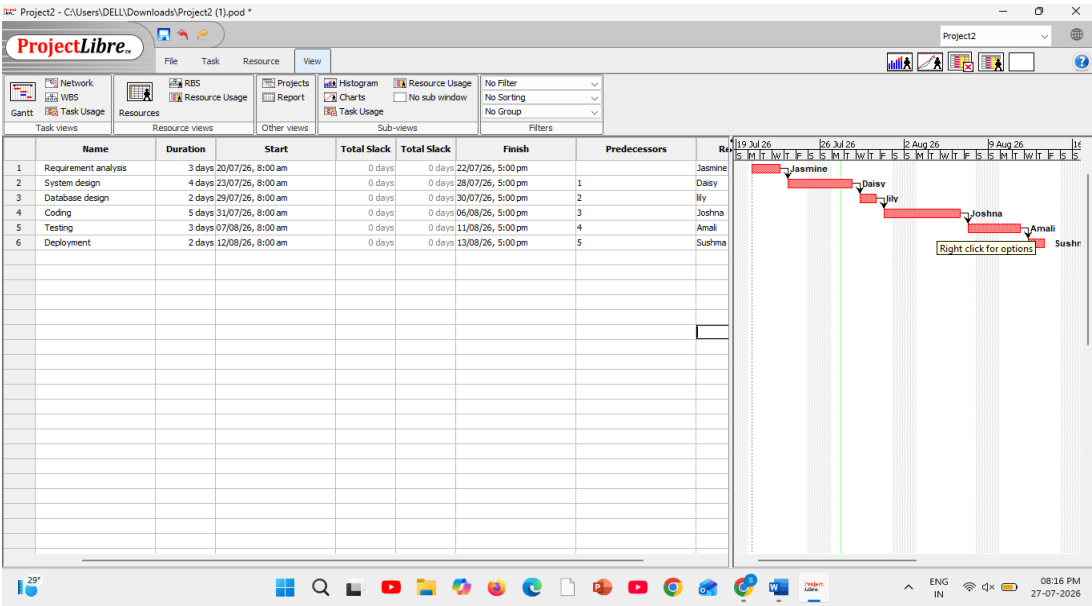
ALGORITHM:

1. List all project activities and their durations.
2. Identify the dependencies between activities.
3. Construct the CPM network diagram.
4. Calculate the Earliest Start (ES) and Earliest Finish (EF) for each activity.
5. Calculate the Latest Start (LS) and Latest Finish (LF) for each activity.
6. Determine the slack (float) for each activity.
7. Identify the critical path and estimate the total project duration.

PROCEDURE:

1. Open ProjectLibre and create a new project.
2. Enter all project activities with their durations.
3. Specify the predecessor for each activity.
4. Assign resources to the activities if required.
5. Generate the Gantt Chart and Network Diagram.
6. Analyze the schedule to identify the critical path and total slack.
7. Save the project and generate the project schedule report.

OUTPUT:



RESULT: